DPP-BO04:10: Inward Turbulence Spreading Across the Core-Edge Transition and Implications for Core Modeling

Approaching the Density Limit

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Themes

Microphysics Focus

- Key dimensionless parameters: **Kubo Number** (Ku) and **Production Ratio** (R_p)
- Micro-Macro connection by experimental scans $\rightarrow n/n_G$
- The Physics of the turbulence spreading phenomenon

Key Questions

- What is the anomalous physics when Ku (a local parameter) exceeds unity?
- When does the contamination/spreading process dominate over local production?
- How is the often-quoted adiabaticity parameter $\alpha = \frac{k_\parallel^2 v_{\mathrm{th,e}}^2}{\nu_{\mathrm{ei}}|\omega|}$ linked to $Ku > 1$ and the spreading phenomenon?

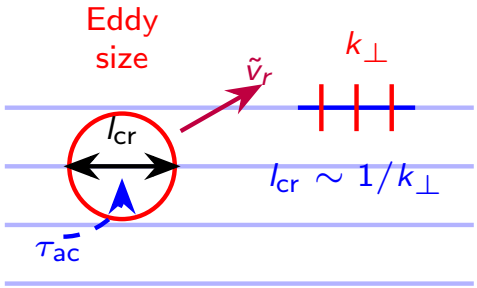
The physics

The Kubo Number Ku

Definition

$$Ku = \frac{\tilde{v}_r \cdot \tau_{ac}}{l_{cr}}, \qquad \text{(R. Kubo, 1963)}$$

- $\tilde{v}_r$: radial velocity fluctuation
- τ_{ac}: auto-correlation time from e-folding time of $R(\tau) = \langle \tilde{n}(t)\tilde{n}(t+\tau)\rangle / \langle \tilde{n}(t)^2 \rangle$
- $l_{cr} = 1/\sigma_k$: radial correlation length estimated from inverse of the spectral width of radial wavenumber spectrum



$Ku < 1$: Diffusive Regime $\rightarrow$
Many kicks in correlation length

$Ku > 1$: Persistent Regime $\rightarrow$
Long τ_{ac}, few kicks

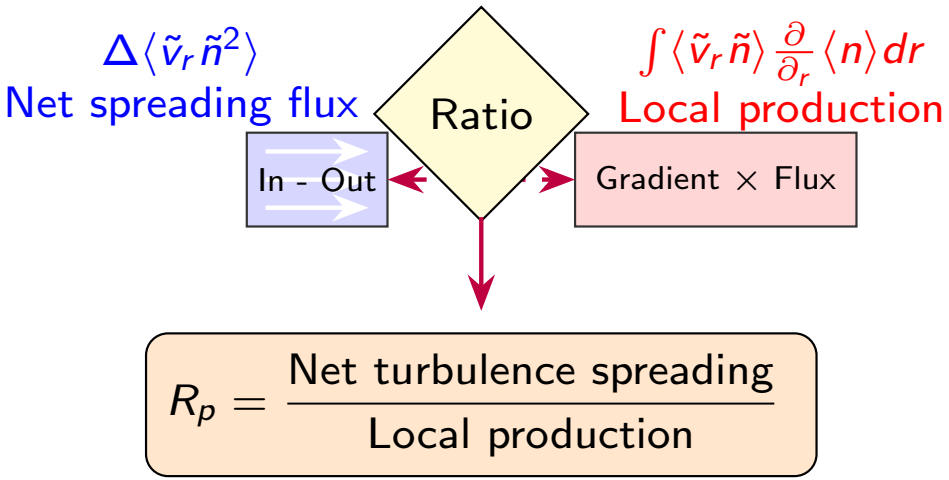
Production Ratio R_p

Definition

The production ratio is the ratio of net spreading drive to net local production in interval. R_p is defined as:

$$R_p = \frac{\Delta\langle \tilde{v}_r \tilde{n}^2 \rangle}{-\int_{r-\delta}^{r+\delta} \langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r}\, dr}, \qquad \text{(c.f. P. Manz, T. Wu, P.D.)}$$

where the interval referenced above is from LCFS to $\max(1/L_p)$.



Adiabaticity α - A Local Parameter

Definition

$$\alpha = \frac{k_{\parallel}^2 v_{\mathrm{th},e}^2}{\nu_{ei}|\omega|}, \quad k_{\parallel} \approx \frac{1}{qR}$$

- $v_{\mathrm{th},e} = \sqrt{K_B T_e / m_e}$: the electron thermal velocity
- ν_{ei}: electron-ion collision frequency
- $|\omega|$: frequency of the drift wave

$\alpha > 1$:

- Adiabatic electron response
- Drift wave turbulence
- Zonal flow generation

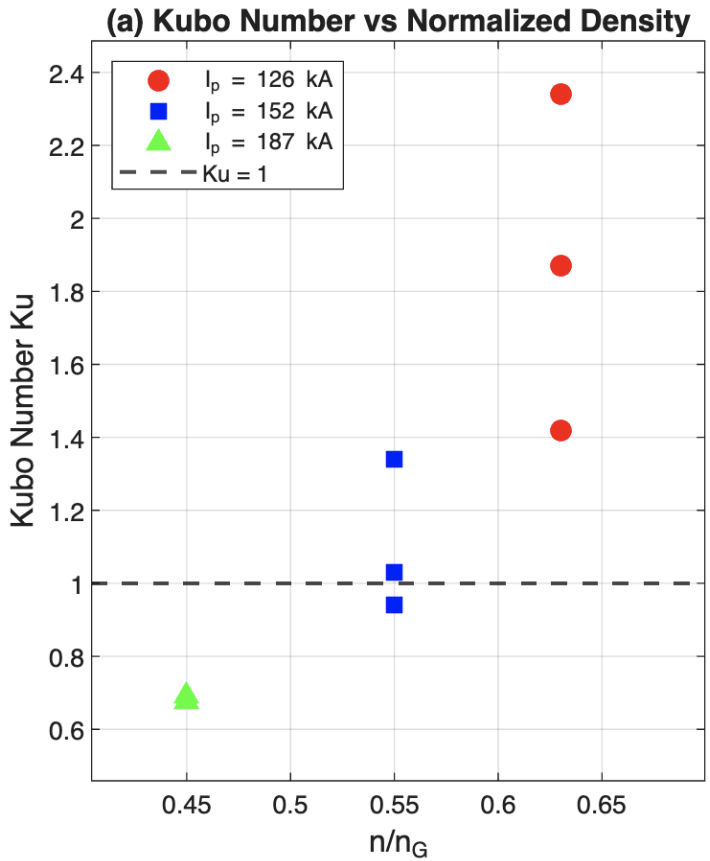
$\alpha \ll 1$:

- Non-adiabatic electron response
- Hydrodynamic turbulence
- Zonal flow collapse

Important: α linked to the density limit (Many authors)

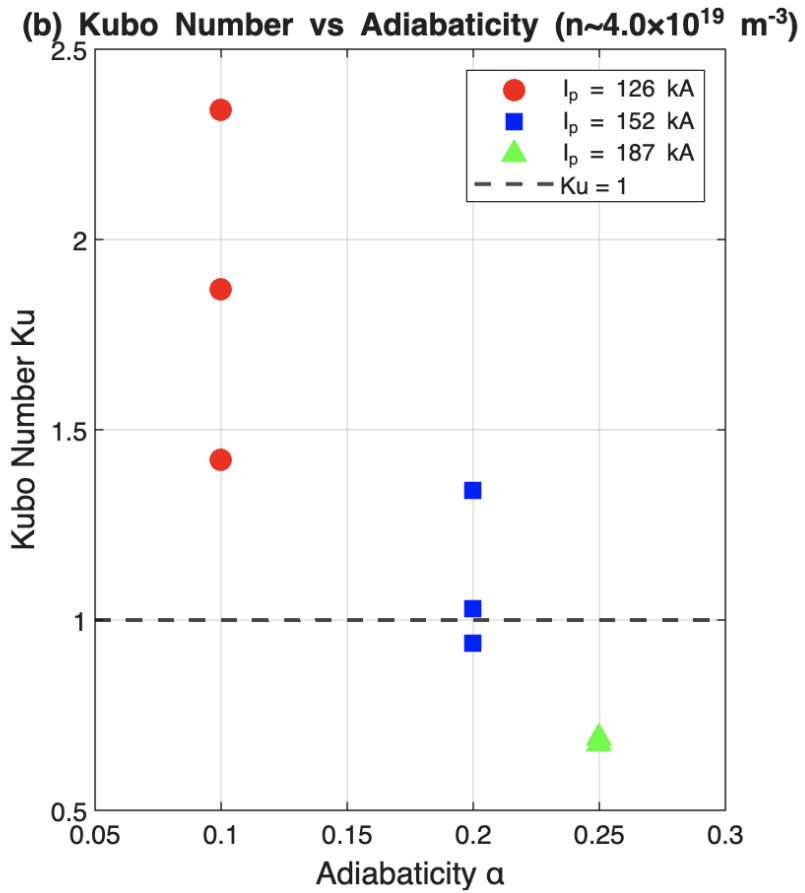
Data & Experiment

Experimental Results: Kubo Number Ku vs n/n_G (Global)



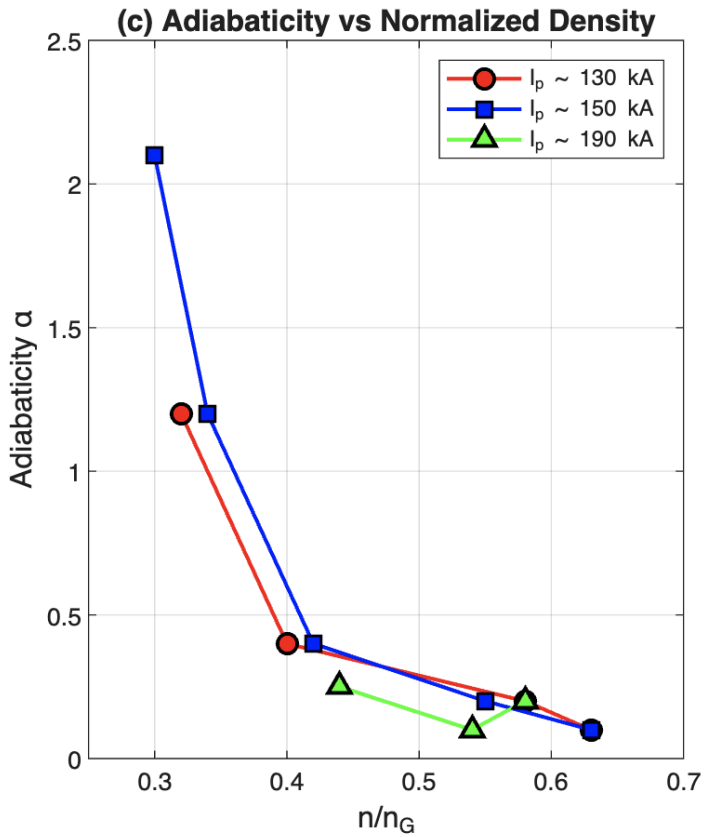
- The Kubo number Ku increases from 0.7 to 2.1 ($Ku > 1$ clear)
- The transition through $Ku = 1$ occurs near $n/n_G \sim 0.5$
- Zonal flow collapse at high n/n_G (R. Hong)

Cont'd: Ku vs α



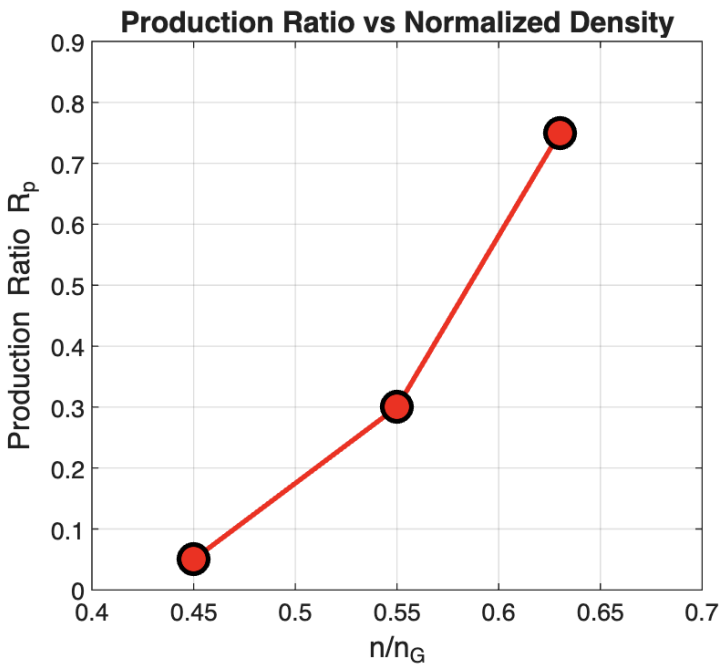
- As α decreases (increased electron-ion collisions), Ku increases
- This reveals an inverse relationship: $Ku \propto 1/\alpha$
- Higher collisionality (lower α) leads to higher τ_{ac} and thus, higher Ku

Cont'd: α vs n/n_G



- Increasing the Greenwald density (i.e., more electron-ion collisions), reduces the adiabaticity parameter α
- A higher current increases the Greenwald density n_G, which lowers n/n_G at fixed density, so α vs I_p is a positive correlation instead

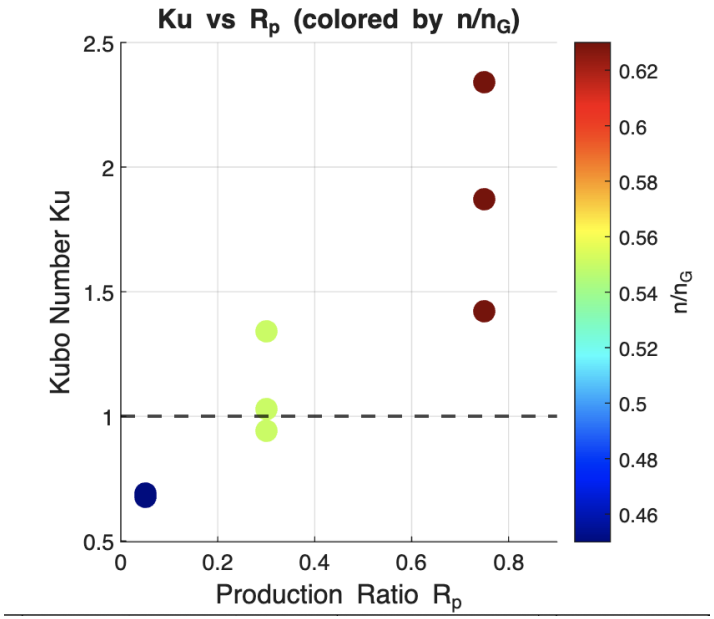
R_p (Production Ratio) vs n/n_G

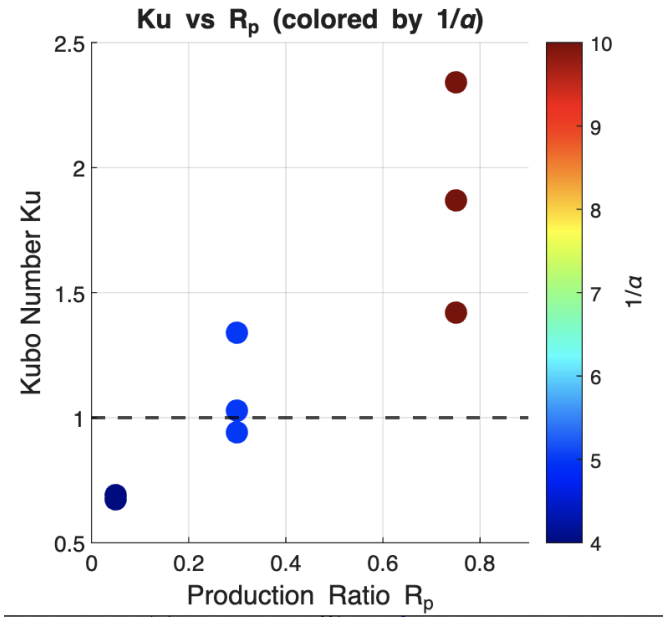


- The production ratio R_p increases from 0.1 to 0.8 as n/n_G increases from 0.5 to 0.7
- At the highest density, R_p approaches unity
- This suggests a transition from a local-production-driven regime to a spreading-significant regime approaching the DL

Key Analysis

Ku vs R_p Colored by n/n_G and $1/\alpha$





- Color coding by n/n_G reveals the density dependence
- Higher density corresponds to higher values of both Ku and R_p

- $1/\alpha$ decreases as R_p and Ku increase
- Lower α indicates an hydrodynamic regime, favoring turbulence spreading

Large Ku Turbulence Spreading $\longleftrightarrow$ Blob Propagation

Blob Kubo Number

The blob Kubo number is defined as:

$$Ku_{\text{blob}} = \frac{v_{\text{prop}} \cdot \tau_{\text{life}}}{l_{\text{pene}}}$$

- v_{prop}: blob propagation speed
- τ_{life}: blob lifetime
- l_{pene}: radial blob penetration depth

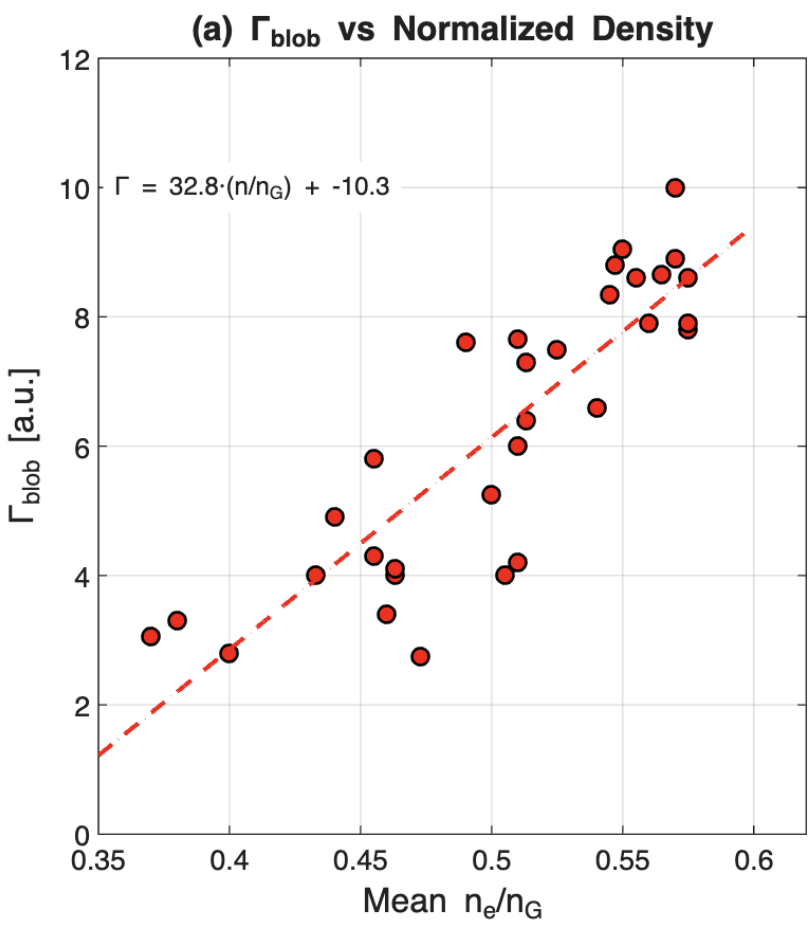
Blob Kubo Number Remains at Unity

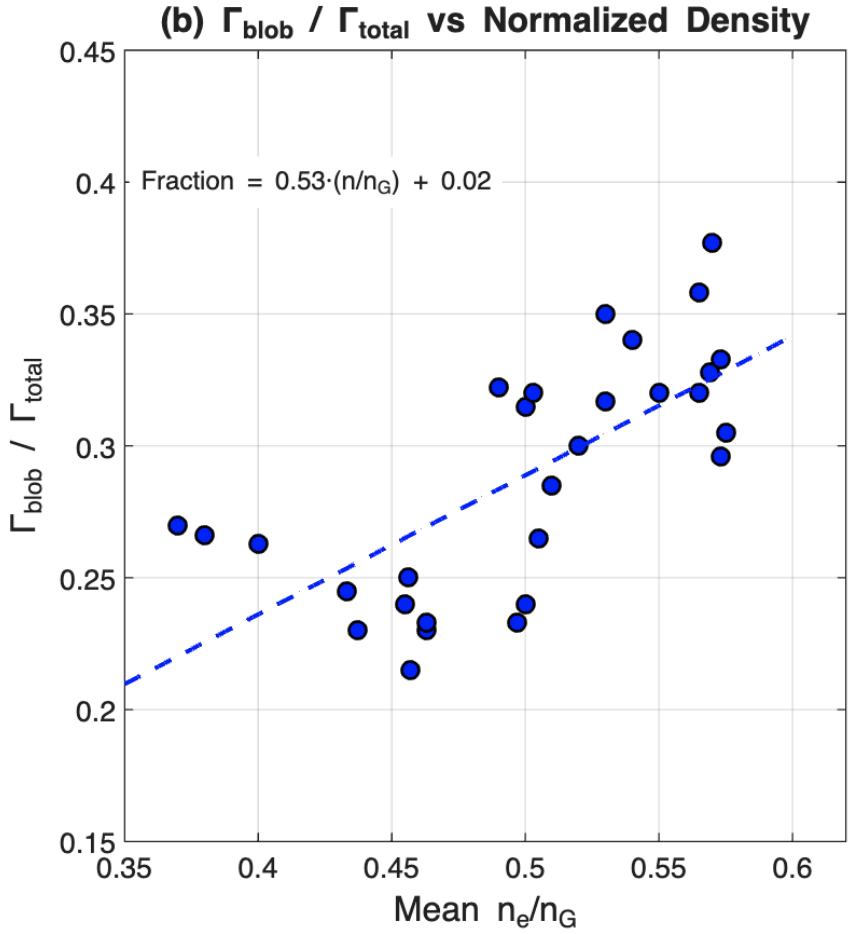
Discharge	**Plasma Current I_p [kA]**	**Adiabaticity α**	**Blob Kubo Number ($K_{u,\mathbf{blob}}$)**
Discharge 1	187	0.25	0.945
Discharge 2	126	0.40	1.050

Table: Blob Kubo number versus plasma current and adiabaticity.

- **At high n/n_G (low α), $Ku_{\mathbf{blob}} \sim 1$ across all measured conditions**
- Blobs self-organize to maintain $v_{\mathrm{prop}}\tau_{\mathrm{life}} \sim l_{\mathrm{pene}}$
- **The duality of $Ku > 1$ and $Ku_{\mathbf{blob}} \gtrsim 1$ for large n/n_G suggests that turbulence $\longleftrightarrow$ a gas of blobs, not "modes"**

Blob Transport Scaling with Normalized Density





- The outward particle transport carried by blobs scales roughly linearly with the normalized Greenwald density
- At low density, blobs effects are minor. But near the Greenwald density limit, blobs dominates the transport

Conclusions

1. **The $Ku \sim R_p$ correlation demonstrates that spreading and contamination/spreading processes underlie $Ku > 1$ phenomena.** Long auto-correlation times and spreading strength are physically linked
2. **Blob spreading** emerges with $Ku_{\text{blob}} \to 1$ at high n/n_G. The blob-induced particle flux approaches 50% of total transport when $Ku > 0.5$, confirming that blobs become more dominant in the interaction-dominated regime
3. The adiabaticity α is a key **local parameter** in determining Ku and R_p
4. **Message for modelers**: At high n/n_G, turbulence is best modeled as a gas of blobs, not as an ensemble of linear modes. Traditional quasi-linear models fail!

Conclusions: Future Aims

1. **Extend to larger devices**: Apply Ku and R_p analysis to HL-3 and DIII-D to verify scaling relationships across machine sizes
2. **Quantify the R_p phase space**: Map the boundary between local-production-dominated ($R_p < 1$) and spreading-dominated ($R_p > 1$) regimes as a function of n/n_G and I_p
3. **Study edge-core coupling**: Investigate how turbulence spreading from the edge contaminates the core and whether core density saturation via avalanching is linked to $R_p > 1$ edge dynamics
4. **Study the relation between R_p and zonal flow**: R_p vs Shearing Kubo number

Thank You!

Production Ratio R_p Continued (BACKUP)

Derivation

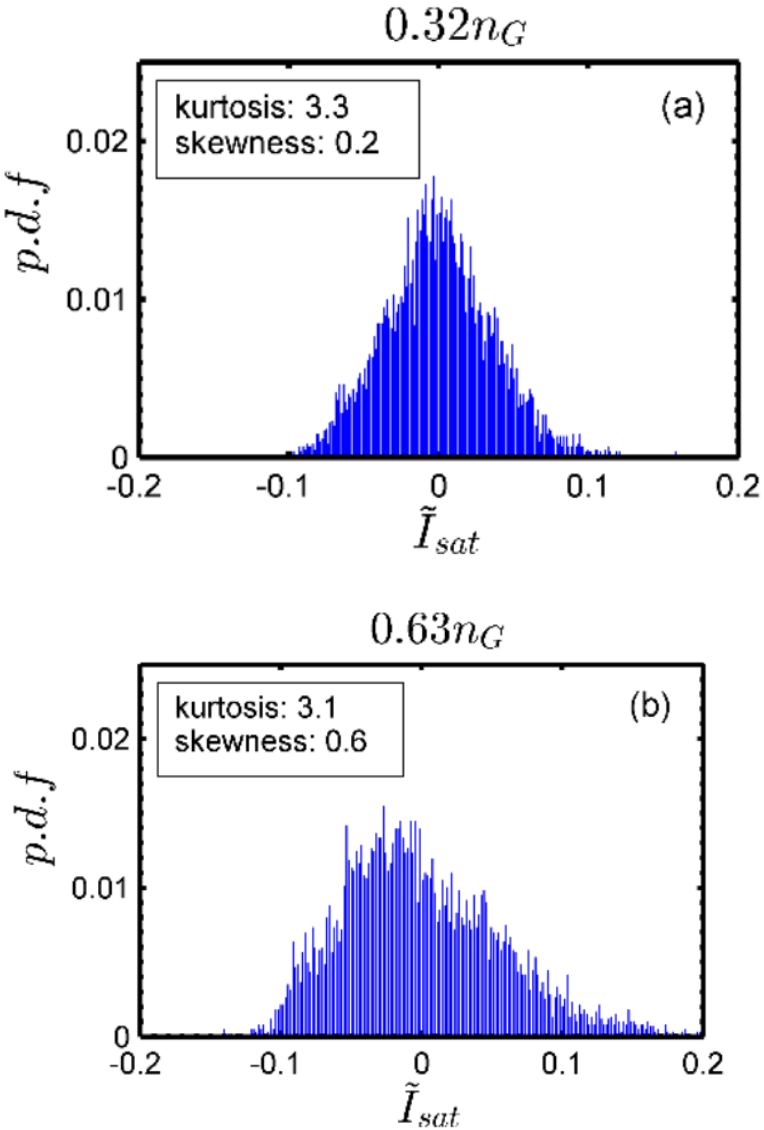
The production ratio emerges naturally from the turbulence intensity evolution equation:

$$\frac{\partial}{\partial t}\left(\frac{\langle \tilde{n}^2 \rangle}{2}\right) = -\langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r} - \frac{\partial}{\partial r}\left(\frac{\langle \tilde{v}_r \tilde{n}^2 \rangle}{2}\right) - \epsilon$$

where the first term on the right is local production, the second term is the turbulence spreading flux, and the third term is the external dissipation sources. R_p is defined as:

$$R_p = \frac{\Delta \langle \tilde{v}_r \tilde{n}^2 \rangle}{-\int_{r-\delta}^{r+\delta} \langle \tilde{v}_r \tilde{n} \rangle \frac{\partial \langle n \rangle}{\partial r}\, dr}$$

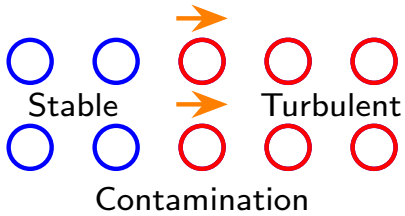
An Interplay of *Ku* and Skewness - T. Long



Key Observations

- As density approaches n_G, skewness also increases $\rightarrow$ Skewness increases with *Ku*
- Positive skewness indicates **blob-dominated** turbulence
- Blobs carry density radially outward

The Contamination Analogy (Pomeau, 1986) (BACKUP)



"Each oscillator if in a turbulent state may either relax spontaneously towards its quiescent state or contaminate its neighbors. - Pomeau, 1986"

Pomeau's Perspectives

- The analysis of reaction-diffusion systems reveals that the interaction between nearby oscillators may be seen as a sort of contamination process
- This is precisely where the turbulent state spreads through interactions rather than local instabilities, analogous to the $R_p < 1$ regime
- The robustness of the spreading front is consistent with the interaction-driven regime becoming dominant near operational limits